Tomita–Takesaki Theory

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Abstract

These notes were typed and reorganized from the final lecture on von Neumann algebras in the *Groundwork for Operator Algebras Lecture Series* at Michigan State University, given by Professor Brent Nelson.

1 The L^2-space of a faithful normal state

Let M be a von Neumann algebra, and let $\varphi : M \longrightarrow \mathbb{C}$ be a faithful normal state.

Definition 1.1. Define

$$\langle x, y\rangle_\varphi = \varphi(y^* x), \qquad x, y \in M.$$

Then

$$L^2(M, \varphi) := \overline{M}^{\|\cdot\|_\varphi}.$$

We regard $M \subset L^2(M, \varphi)$, and write $\hat{x}$ for the image of $x \in M$ in $L^2(M, \varphi)$.

We can represent $M \subset B(L^2(M, \varphi))$ by $x \cdot \hat{y} = \widehat{xy}$, and this action extends to all of $L^2(M, \varphi)$. Recall that when $\varphi = \tau$ is a trace, the map $\hat{x} \longmapsto \widehat{x^*}$ is a conjugate-linear isometry. Indeed,

$$\|\widehat{x^*}\|_\tau^2 = \tau(xx^*) = \tau(x^*x) = \|\hat{x}\|_\tau^2.$$

2 An unboundedness example

Example 2.1. Consider $M = B(H)$, with H infinite-dimensional, and let φ be a faithful normal state on $B(H)$.

Recall that there exists $a \in L^1(B(H))$ such that

$$\mathrm{Tr}(a) = 1, \qquad \varphi(x) = \mathrm{Tr}(ax) \quad \forall x \in B(H).$$

Since φ is positive, $a \geq 0$. Since φ is faithful, $\ker a = 0$.

By the spectral theorem for compact normal operators, there exist pairwise orthogonal projections $\{p_n : n \in \mathbb{N}\}$ and numbers $\{\lambda_n > 0 : n \in \mathbb{N}\}$ such that

$$\sum_{n\geq 1} p_n = 1 \quad \text{(strongly)}, \qquad a = \sum_{n \geq 1} \lambda_n p_n.$$

Moreover, each p_n is finite rank; after grouping equal eigenvalues, assume $\lambda_1 > \lambda_2 > \cdots$. We have $\lambda_n \underset{n\to\infty}{\longrightarrow} 0$.

Choose a rank-one projection $q_1 \le p_1$; for each $n \ge 2$, choose a rank-one projection $q_n \le p_n$. Since $B(H)$ is a factor, no two nonzero projections are centrally orthogonal. Taking q_1 and q_n rank one, we have $q_1 \sim q_n$. Hence there exists $v_n \in B(H)$ such that $v_n v_n^* = q_1$, $v_n^* v_n = q_n$. Then

$$\begin{aligned}\|\widehat{v_n^*}\|_\varphi^2 &= \varphi(v_n v_n^*) = \varphi(q_1) = \mathrm{Tr}(a q_1)\\ &= \mathrm{Tr}(\lambda_1 q_1) = \lambda_1 \dim(q_1 H).\end{aligned}$$

On the other hand,

$$\|\widehat{v_n}\|_\varphi^2 = \varphi(v_n^* v_n) = \varphi(q_n) = \mathrm{Tr}(\lambda_n q_n) = \lambda_n \dim(q_n H).$$

So, using $q_1 \sim q_n$,

$$\frac{\|\widehat{v_n^*}\|_\varphi}{\|\widehat{v_n}\|_\varphi} = \frac{\sqrt{\lambda_1}}{\sqrt{\lambda_n}} \underset{n\to\infty}{\longrightarrow} +\infty.$$

Thus the map $\hat{x} \mapsto \widehat{x^*}$ need not be bounded.

3 The Tomita operator and modular objects

Definition 3.1. Define the conjugate-linear operator

$$S_\varphi : \hat{x} \longmapsto \widehat{x^*}$$

on the dense subspace $M \subset L^2(M, \varphi)$.

While S_φ can be unbounded, it is still densely defined; in fact, it is closable. Denote its closure again by S_φ. Then S_φ admits a polar decomposition

$$S_\varphi = J_\varphi \Delta_\varphi^{1/2} = \Delta_\varphi^{-1/2} J_\varphi, \qquad \Delta_\varphi = S_\varphi^* S_\varphi.$$

Here J_φ is a conjugate-linear isometry such that $J_\varphi^2 = 1$, $J_\varphi = J_\varphi^*$; it is called the modular conjugation. The operator Δ_φ is positive, non-singular, and self-adjoint; it is called the modular operator.

4 Miracles

Theorem 4.1 (Tomita–Takesaki)**.** *The main miracles are:*

(i)
$$J_\varphi M J_\varphi = M' \cap B(L^2(M, \varphi)).$$

(ii) *For every $t \in \mathbb{R}$,*
$$\underbrace{\Delta_\varphi^{it}}_{unitary} M \Delta_\varphi^{-it} = M.$$

5 Examples of the modular operator

(1) The state φ is tracial if and only if $\Delta_\varphi = 1$.

(2) Let $M = B(H)$, let $\varphi = \mathrm{Tr}(a\,\cdot)$, and write

$$a = \sum_{n\geq 1} \lambda_n p_n.$$

If $x \in B(H)$ satisfies $p_n x p_m = 0$ for all but finitely many $(m,n) \in \mathbb{N}^2$, then

$$\Delta_\varphi \hat{x} = \widehat{axa^{-1}},$$

where a^{-1} is understood on the finite spectral support of x. Indeed, for such x, and for y satisfying the same finite-support condition,

$$\begin{aligned}
\langle \Delta_\varphi \hat{x}, \hat{y}\rangle_\varphi &= \langle S_\varphi \hat{y}, S_\varphi \hat{x}\rangle_\varphi \\
&= \langle \widehat{y^*}, \widehat{x^*}\rangle_\varphi \\
&= \varphi(xy^*) = \mathrm{Tr}(axy^*) \\
&= \mathrm{Tr}(y^*ax) = \mathrm{Tr}(a^{-1}ay^*ax) \\
&= \mathrm{Tr}(ay^*axa^{-1}) \\
&= \varphi(y^*axa^{-1}) = \langle \widehat{axa^{-1}}, \hat{y}\rangle_\varphi.
\end{aligned}$$

Hence $\Delta_\varphi \hat{x} = \widehat{axa^{-1}}$ under this finite-support condition.

(3) Recall that if Γ is a countable discrete group, then $L^2(L(\Gamma), \tau) = \ell^2(\Gamma)$.

Using the same construction for faithful normal semifinite weights, let G be a locally compact group with left Haar measure μ. There exists a "Plancherel weight" φ on $L(G)$ such that

$$L^2(L(G), \varphi) = L^2(G, \mu).$$

The modular operator Δ_φ is pointwise multiplication by the modular function $\Delta_G : G \longrightarrow \mathbb{R}_+$:

$$(\Delta_\varphi \xi)(s) = \Delta_G(s)\xi(s).$$

Here we use the convention

$$\mu(E \cdot s) = \Delta_G(s)\mu(E)$$

for every Borel set $E \subset G$.

6 The modular automorphism group and crossed product $c(M)$

Observe that, by the Tomita–Takesaki theorem, $\sigma_t^\varphi(x) := \Delta_\varphi^{it} x \Delta_\varphi^{-it}$ defines a $*$-isomorphism of M onto M. The one-parameter group $\sigma^\varphi = (\sigma_t^\varphi)_{t\in\mathbb{R}}$ is the modular automorphism group. In fact, we obtain an action

$$\mathbb{R} \overset{\sigma^\varphi}{\curvearrowright} M.$$

For a chosen faithful normal state φ,

$$c(M) := M \rtimes_{\sigma^\varphi} \mathbb{R}.$$

Theorem 6.1 (Connes–Takesaki)**.** *The crossed product* $c(M)$ *does not depend, up to isomorphism, on the choice of* φ*.*

If M *is type* III*, then* $c(M)$ *is semifinite, and* M *is a factor if and only if the dual action*

$$\mathbb{R} \overset{\theta}{\curvearrowright} c(M)$$

restricts to an ergodic action on $Z(c(M))$*.*

Theorem 6.2 (Takesaki duality)**.** *After identifying* $\widehat{\mathbb{R}} \cong \mathbb{R}$*, let* $\mathbb{R} \overset{\theta}{\curvearrowright} c(M)$ *denote the dual action. Then*

$$c(M) \rtimes_\theta \mathbb{R} \cong M \mathbin{\bar{\otimes}} B(H), \qquad H = L^2(\mathbb{R}).$$

References

[1] S. J. Summers, *Tomita–Takesaki modular theory*, in *Encyclopedia of Mathematical Physics*, vol. 5, J.-P. Françoise, G. L. Naber, and T. S. Tsun, eds., Elsevier, 2006, pp. 251–257. arXiv:math-ph/0511034.